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Ecological inference and US elections

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Abstract

This thesis investigates the problem of ecological inference, which arises when we are interested in relationships between variables at the individual level but only have access to aggregate data.

Section 1 presents the problem of ecological inference, its relevance in various contexts and some of the solutions proposed. In particular, it presents the method proposed by McCartan and Kuriwaki (2025), which is the one that will be used in this thesis.

Section presents a synthetic example to illustrate the method and its performance in a controlled setting, where the true individual-level relationships are known.

Section presents an application of the method to US elections, where we are interested in the relationship between individual-level characteristics and individual voting behavior, but we only have access to aggregate data.

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1 Introduction

Ecological inference fallacy is the problem that arises when we are interested in relationships between variables at the individual level but we only observe data at the aggregate level.

This problem is relevant in all situations in which data are reported at the aggregate level, but we are interested in relationships at the individual level.

A famous example of that problem is the one presented by Robinson (1950) for the relationship between illiteracy and race, and illiteracy and migration in the United States. In that case, the author observed that areas with higher proportion of Black population, as well as areas with higher proportion of migrants, had lower illiteracy rates. However, when looking at the individual level, Black people and migrants were more likely to be illiterate than White and non-migrant people. The explanation of this apparent paradox is that Black and migrant people were more likely to live or move to areas with lower illiteracy rates, and therefore the aggregate data masked the individual-level relationship.

We can think also at economic examples. Consider, for instance, a retail store chain that wants to investigate consumers' habits of young people observing data about sells in different stores. The retail store chain might know quantities of goods sold in different stores and, at the same time, demographic data about consumers in different areas covered by different stores. The naive researcher, therefore, would link the percentage of young people and the quantities of different goods sold, suggesting that goods sold in stores with younger consumers are sold to young consumers. However, that is a clear example of ecological fallacy and it could lead to false findings. In fact, when we consider data about cities, it might well be that stores with greater share of young consumers are located in central areas near universities, where richer people also usually live.

Therefore, even if young people might be cheaper consumers, their habits would be masked by their neighbours' habits.

In any case, the most obvious case of ecological inference is in electoral behavior, where we are interested in the relationship between individual-level characteristics and individual voting behavior, but obviously (and luckily!) we are not able to observe individual-level voting behavior, and sometimes we are not able to observe individual-level characteristics either. In these cases, we are forced to rely on aggregate data, possibly choosing the most granular level of aggregation possible. Unfortunately, however, even choosing the most granular level of aggregation possible, individual-level relationships may be severely masked. That, of course, is highly relevant for political economists, who often times deal with the relationships between demographic characteristics, electoral results and public policies.

Kuriwaki and McCartan (2026) present a particularly convincing example of that problem, citing 1968 US presidential elections, where, in addition to the Republican and Democratic candidates Richard Nixon and Hubert Humphrey, there was also a relevant third-party candidate, George Wallace, who lead the segregationist American Independent Party. George Wallace had his strongest results in the Southern States, where the proportion of Black population was generally higher than in the rest of the nation. Therefore, a very naive analysis of the aggregate data would suggest that Black people were more likely to vote for George Wallace, whose platform was segregationist and overtly racist. Kuriwaki and McCartan (2026) go on to show that, even at more granular levels of aggregation, such as the county level, the relationship between proportion of Black population and proportion of votes for George Wallace would still be positive considering all counties in the nation; not only that, but even restricting the analysis to some specific States, such as North Carolina, we would still observe a positive relationship: the higher the proportion of Black population, the higher the proportion of votes for George Wallace.

The key here is that, while Black people were a quite homogenous group in terms of voting behavior (generally favoring the Democratic candidate and certainly not voting for George Wallace), White people were a much more heterogeneous group, and in particular, White people with segregationist views were also more likely to live in areas with higher proportion of Black population.

In fact, that same mechanism is at work in the retail store example: while young consumers might be quite homogeneously cheap in their habits, the prevalent majority of older people has different habits and it drives overall difference.

Robinson (1950) came to the conclusion that ecological fallacy would constitute an inextricable problem and he concluded his paper telling that he was

... aware that this conclusion has serious consequences, and that its effect appears wholly negative because it throws serious doubt upon the validity of a number of important studies made in recent years. The purpose of this paper will have been accomplished, however, if it prevents the future computation of meaningless correlations and stimulates the study of similar problems with the use of meaningful correlations between the properties of individuals.

Econometrics in 1950, however, had considerably fewer instruments than it has now and we might think that some of these techniques could be successfully deployed in confronting ecological inference fallacy.

In particular, we could be tempted to argue that the issue could be solved by using covariates.

That, however, is not necessarily sufficient. In fact, when Kuriwaki and McCartan (2026) look at results in presidential elections in North Carolina, they are implicitly adding a covariate for the State and interacting it with the proportion of Black voters. Indeed, the estimate they present can be thought

as the coefficient associated to the share of Black voters, given the covariate of the county being in North Carolina.

Analogously, we might include each possible interaction between variables and functional forms, for example imposing a non-linear relationship between Black voters' share and electoral results; that could result in a positive relationship between Black voters' share and segregationist votes for relatively small shares of Black voters that would later reverse when Black voters become majority.

Another method that we could possibly consider is the inclusion of fixed effects. However, fixed effects require repeated observations of the same unit, which is not the case for our examples.

All these examples highlight the common problem in adopting traditional econometric techniques to address ecological inference, which is the difference between the unit of observation and the quantity that we want to estimate.

In fact, we can think of each possible technique to better identify the relationship between variables at the level of unit of observation, but that would not solve the ecological fallacy of composition.

Indeed, even our naivest estimates might be correct in explaining relationships at the level of units of observations. It can be true that counties with more migrants have lower illiteracy rates even accounting for other variables or other effects, or that consumers in areas with younger people spend more in expensive goods, or that precincts with higher shares of Black voters favor segregationist candidates.

What these data can't tell us is whether migrants have lower or higher illiteracy rates, younger consumers spend more or less in expensive goods, or how Black people vote.

In order to answer these questions, we have to recover a notation that explicitly

accounts for the aggregate nature of our units of observation and to impose a set of assumptions.

Only that will allow us to move from the agnostic conclusion of Robinson (1950) to ecological estimates.

In the next Section, I will introduce a general notation for ecological inference problems and I will briefly present some of the methods employed to address ecological inference and the set of assumptions they rely on.

In the following Section, I will present a more specific model for electoral analysis.

Both of these Sections will be heavily dependent on Kuriwaki and McCartan (2026), where the authors focus on the assessment of ecological inference methods, and McCartan and Kuriwaki (2025), where the authors present the semiparametric method called *seine*. In any case, the notation that I will present is slightly different and more apt to describe the issues that we will face in this dissertation.

Once the model is presented, I will present a synthetic example to show how the method works and to assess its performance in a controlled environment.

Finally, I will present an application to real data about US presidential elections and I will conclude with some comments and conclusions.

2 Notation for ecological inference

We consider a population of N individuals, that we denote by $i \in [1, N]$.

Each individual i is characterized by a set of variables, which we can divide into a response variable Y_i and a categorical variable X_i .

The response variable can be either quantitative or categorical, and it will be denoted consequently. In case of a quantitative variable, we will denote it by a scalar $Y_i \in \mathbb{R}$. In case of a categorical variable, we will denote it by a vector $Y_i \in \{0, 1\}^p$, where p is the number of possible categories: that is, Y_i is a vector with all entries equal to 0 except for the entry corresponding to the value of the categorical variable for individual i , which is equal to 1.

That means that $Y_i = (1, 0, 0, \dots, 0)$ if the response variable is the first category, $Y_i = (0, 1, 0, \dots, 0)$ if the response variable is the second category, and so on...

The categorical variable X_i , instead, is consequently denoted by $X_i \in \{0, 1\}^d$, where d is the number of possible categories, and it encodes which is the category to which individual i belongs. We shall denote the categories with $k \in [1, d]$.

The total population is partitioned into G geographies, which we denote by $g \in [1, G]$: each individual belongs to one and only one geography, which we denote by G_i .

Geographies are what we previously referred to as units of observation, therefore we are never able to observe Y_i and X_i , but we only observe variables at geography level.

For each geography g , we have population $N_g = \sum_i \{1|G_i = g\}$.

Moreover, we have, $\bar{Y}_g = \frac{1}{N_g} \sum_i \{Y_i|G_i = g\}$ and $\bar{X}_g = \frac{1}{N_g} \sum_i \{X_i|G_i = g\}$.

$\bar{Y}_g$ can be either a scalar, if the response variable is quantitative, or a vector of shares $\bar{Y}_g \in [0, 1]^d$, if the response variable is categorical; in this second case, each entry of the vector would represent the share of the respective category and the sum of entries would be 1 as the categories are a partition of the population.

$\bar{X}_g$, instead, is a vector of shares $\bar{X}_g \in [0, 1]^d$.

2.1 Global and local estimands

In general, we are interested in the average of Y_i conditional on the belonging to category $k \in [1, d]$.

We shall denote this quantity by $B_k = \frac{\sum_i \{Y_i|i \in I_k\}}{N_k}$, where I_k is the subset of people belonging to category k , while N_k is the cardinality of I_k .

As before, B_k is a scalar if the response variable is quantitative and a vector of shares if the response variable is categorical.

When we consider all different categories $k \in [1, d]$ we end up either with a vector $\beta \in \mathbb{R}^d$ or a matrix $\beta \in [0, 1]^{p \times d}$.

We shall call β the global estimand.

However, as always, we do not observe individuals and therefore it is useful to recover quantities and definitions at geographical level.

The local estimand for category k and geography g is defined as $B_{gk} = \frac{\sum_i \{Y_i|i \in I_{gk}\}}{N_{gk}}$, where I_{gk} is the subset of people belonging to category k and geography g and N_{gk} is its cardinality.

Analogously, we can collect all B_{gk} pertaining to the same geography g but to different categories k within β_g of dimension d or $p \times d$.

These definitions allow us to derive two accounting identities.

The first one links global estimand to local estimands: $\beta = \frac{\sum_{g=1}^G \beta_g * N_g}{N}$, i.e. the global estimand is the weighted average of local estimands, weighted by the population of each geography.

Therefore, this identity suggests a strategy that starts from local estimands, which can also be of interest on their own, to recover the global estimand.

The second accounting identity is at geographical level: $\bar{Y}_g = \beta'_g \bar{X}_g$.

One consequence of this second identity is that each observation contributes d or $p \times d$ parameters, making the local problem unidentifiable.

2.2 Different assumptions, different strategies

2.2.1 CCAR assumption

The easiest way to escape the identification issue is to assume that $\beta_g = \beta, \forall g$ and, consequently, adopt a simple linear regression.

This assumption, however, is particularly stringent and it can be easily falsified by data, for example by finding two geographies with same $\bar{X}_g$ but different $\bar{Y}_g$.

A slightly more relaxed assumption is allowing for variations in β_g , but with the constraint that β_g are independent of $\bar{X}_g$: $E[\beta_g | \bar{X}_g] = \beta, \forall g$; or, equivalently, $E[(\beta_g - \beta) | \bar{X}_g] = 0, \forall g$.

That assumption, which cannot be falsified by observable data per se, would still allow us to use linear regression.

In fact, considering geographies g as units of observations and $\bar{Y}_g$ and $\bar{X}_g$, it would be equivalent to a traditional linear model of the following form: $\bar{Y}_g = \beta_g * \bar{X}_g = \beta * \bar{X}_g + \epsilon_g$, with $\epsilon_g = (\beta_g - \beta) * \bar{X}_g$.

Thanks to the assumption that $E[(\beta_g - \beta)|\bar{X}_g] = 0$, we would have $E[\epsilon_g|\bar{X}_g] = E[(\beta_g - \beta) * \bar{X}_g|\bar{X}_g] = E[(\beta_g - \beta)|\bar{X}_g] * \bar{X}_g = 0$ and linear regression could be used.

As underlined by Kuriwaki and McCartan (2026), that is what Goodman (1953) described, even if with a more primitive notation, when discussing conditions that would allow to exploit linear regressions in ecological inference, bearing in mind Robinson (1950)'s worries.

This assumption is called Coarsening Completely At Random, as it is equivalent to think that the way in which individuals are divided into geographies is independent of other characteristics of the geography: literate and illiterate White people, cheaper and more expensive older consumers are randomly assigned to counties and retail stores.

Equivalently to linear regression, CCAR would allow the following identification: $\beta = E[\bar{X}\bar{X}']^{-1}E[\bar{X}'\bar{Y}]$.

In any case, when performing ecological regression, it is hardly the case that this assumption holds true: in the examples presented in the Introduction, that would mean that White and non-migrants people shared the same average illiteracy rates independently on the percentage of Black and migrant people in their county, that older consumers shared spending habits independently on the share of young people in their areas and that White shared the same voting behaviors independently on the percentage of Black people in their precinct.

Kuriwaki and McCartan (2026) put that in the following way:

Is coarsening completely at random plausible in practice? Consider again the 1968 election example. In this case, CCAR means that the preference of White voters for Wallace is unrelated to the proportion of Black voters in a county or the population of the county. For instance, this means that all of the following groups support Wallace to the same extent in North Carolina in expectation: White voters in rural lowland counties with a higher proportion of Black voters, White voters in urban counties, and White voters in mountainous counties with few racial minorities. CCAR is implausible in this setting, and is incompatible with the social scientist's interest in how different individuals sort into different coarsened geographies.

2.2.2 Adding covariates: CAR assumption

What we have not discussed so far is the possibility of adding covariates to the model, which would allow to relax the CCAR assumption.

We shall now introduce a set of covariates Z_g at geography level. As geographies are our units of observation, covariates can either derive from aggregation of individual characteristics or be characteristics of the geography itself: this distinction is not relevant for our purposes, and we will later see examples of both types. A consequence on notation is that we denote covariates as Z_g without the overline, as they are not necessarily averages of individual characteristics.

Thanks to covariates, we are now able to relax the CCAR assumption and considers the following assumption instead: $E[\beta_g | \overline{X}_g, Z_g] = E[\beta_g | Z_g]$.

This assumption is called Coarsening At Random (CAR) and it is equivalent to think that the way in which individuals are divided into geographies is independent of other characteristics of the geography, conditional on covariates: literature

and illiterate White people, cheaper and more expensive older consumers are randomly assigned to counties and retail stores, conditional on covariates.

As a consequence, geographies may have different β_g and, contrary to CCAR, $\bar{X}_g$ can affect β_g , but we assume that there exist covariates Z_g which are able to capture the effect of $\bar{X}_g$ on β_g .

Therefore, we are able to write β_g as a function of the covariates: $\beta_g = E[\beta_g|\bar{X}_g, Z_g] + \epsilon_g = E[\beta_g|Z_g] + \epsilon_g$ with $E[\epsilon_g|\bar{X}_g] = 0$.

Without loss of generality, we can express $E[\beta_g|Z_g]$ with a functional form: $E[\beta_g|Z_g] = f(Z_g)$.

Then, thanks to the second accounting identity, we have a model that links $\bar{Y}_g$ to $\bar{X}_g$ and Z_g : $\bar{Y}_g = \beta_g \bar{X}_g = f(Z_g)' \bar{X}_g + \epsilon_g' \bar{X}_g$.

Taking the conditional expectations on both sides, we have $E[\bar{Y}_g|\bar{X}_g, Z_g] = f(Z_g)' \bar{X}_g$.

3 A model for ecological inference: presenting seine

Here we introduce the model in which Y is a binary variable and therefore we can convert it into a one-dimensional variable that takes 1 or 0 for each individual, and shares in geographies.

Then X is the previously presented categorical variable and Z is a set of geographical covariates.

We assume that the relationship is as presented in the previous section.

4 A synthetic example

In this section I will show a synthetic example that helps explaining why the problem is relevant and why the solution adopted by McCartan and Kuriwaki (2025) helps us in solving it better than naive methods.

In the synthetic example we are the ones who generate the data and therefore we know the exact model of the reality. We show that, if we generate the data and then we try to estimate with different methods we obtain different results and the naive methods do not work well.

In particular, we can exploit the constitutional referendum in Italy to show a possible way in which the naive estimators could not work and therefore we can generate a dataset that encompasses the problem.

The problem is the following: we want to estimate how did voters with different levels of income vote.

If we look at data at regional level, we would be tempted to say that richest people voted more in favour of the government proposal. In fact, the three regions where the votes in favor were more than the votes in contrast (Lombardy, Veneto and Friuli-Venezia Giulia) are among the richest regions in Italy. On the contrary, the regions where No got more than 60% are among the poorest regions of Italy (Basilicata, Campania, Sicily).

However, once we look at data at municipal level, we find that, within the regions, the relationship is nearly inverse: No votes were more frequent in urban areas and other municipalities where income is relatively higher.

That divergence makes us think that if we were able to observe results at individual level, we could find even different relationships between income and voting behavior.

Therefore, we can now generate a synthetic dataset that presents a structure that resembles some stylized facts that we observe in the constitutional referendum results and other notions that we have about electoral behaviors in Italy.

Our synthetic experiment is constructed as follows.

We assume that election takes place in a Country which has three different regions: North, Centre and South.

Each Country is divided in 100 precincts, so that in the end we observe 300 precincts: 100 precincts in the North region, 100 in the Centre region, 100 in the South region.

For each electoral precinct, we observe the percentage of high-income and low-income voters and the electoral results.

Within each income category, there is high variability in their electoral behaviour.

In particular, we impose that high-income people in the North and South precincts have a high propensity of voting in favor of the reform, while in the Centre this percentage is much lower. As for low-income people, we impose that in North they have high propensity of voting in favor, while in the Centre and in the South this propensity is much lower.

At the same time, we impose that the percentage of high-income voters is highest in the North and lowest in the South.

5 An application to US data

An application to US data

6 Other Comments

Other comments adesso facciamo una prova

7 Conclusions

Conclusions

8 References

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